

Zehao Liu

Data Engineer • Machine Learning Engineer • Artificial Intelligence Engineer • Data Scientist • Data Analyst
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Professional Summary

An early-career technical professional with a computer science and mathematics foundation, targeting data engineering, machine learning, AI, and analytics roles with evidence from pipelines, modeling, and delivery work.

- Target roles: Data Engineer, Machine Learning Engineer, Artificial Intelligence Engineer, Data Scientist, and Data Analyst.
- Continuously strengthening data pipeline, ML, and analytics capability through graduate study and public projects.
- Project evidence centers on reliable data delivery, model evaluation, and clear communication of results.

Education

The University of Sydney

Master of Computer Science

Data Science and AI stream, Sydney, Australia

Feb 2025 - Present

- Data pipelines, data engineering, NLP, machine learning, and data analysis / modeling.

Wilfrid Laurier University

Honours Bachelor of Computer Science and Mathematics

Waterloo, Canada

Sep 2019 - Jun 2024

- Algorithms, proofs, statistics, databases, AI, and network security.

Projects

NEM Monitoring Dashboard | [GitHub](#) | [Demo](#)

Data Engineering

- I designed and implemented the ingestion, publishing, and dashboard layers, then documented the deployment and fallback behavior.
- The dashboard presents live facility updates with stable local and Render deployment paths, plus a fallback replay mode when the stream is stale or unavailable.

Infinite Canvas Studio | [GitHub](#) | [Demo](#)

Software Engineering

- I contributed across product scoping, frontend implementation, collaboration workflow, documentation, and deployment.
- The app supports temporary online rooms for collaboration, separate frontend and relay deployments, and export paths for JSON, single-file HTML, and PROJ folder formats.

Medical Image ML Algorithm | [GitHub](#)

AI / ML

- I designed the end-to-end pipeline, implemented the CLI and model training code, added Grad-CAM and error analysis, and documented reproducible setup for local and Apple Silicon runs.
- The CNN reached 90.1% test accuracy, outperforming PCA + Random Forest (65.7%) and PCA + MLP (69.3%). Grad-CAM and confusion-matrix analysis highlighted where spatial features helped and which tissue classes were most often confused.

EduAttain Prediction | [GitHub](#)

Data Analytics

- I contributed to data ingestion and cleaning against ABS metadata, exploratory analysis, model benchmarking, and the Python automation for XGBoost, LightGBM, and CatBoost tuning.
- The workflow produced a cleaned modeling dataset, exploratory reports, and tuned XGBoost, LightGBM, and CatBoost pipelines alongside R baselines such as Random Forest and SVM for highest-attainment classification.

Skills

Technologies: caret, CatBoost, Docker, ggplot2, GitHub Actions, JavaScript, Konva.js, LightGBM, Matplotlib, MQTT, Node.js, NumPy, Pandas, Python, R, Render, Scikit-learn, Streamlit, TensorFlow / Keras, Vercel, Vite, WebSocket, XGBoost

Professional Strengths

- I combine engineering and analytical perspectives, which helps me connect implementation work to business goals and measurable outcomes.
- I can break down complex tasks quickly and use prioritization plus risk control to improve delivery certainty and quality.
- I use AI-assisted coding, research, documentation, and debugging effectively while still validating outputs and keeping them explainable.
- I have cross-functional collaboration experience and can support documentation, reporting, and technical communication.
- I stay steady under fast-paced or high-pressure conditions, balancing detail quality with response speed to keep key work on track.